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Mastering Model Predictive Control (MPC) in Industrial Processes Using MATLAB Simulink

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Introduction

Introduction to Model Predictive Control (MPC)

Model-based predictive control (MPC) describes a set of advanced control methods, which make use of a process model to predict the future behavior of the controlled system. By solving a—potentially constrained—optimization problem.

Unlike traditional reactive control approaches like PID, MPC takes a proactive stance, planning to navigate constraints and optimize performance over a prediction horizon.

Traditional controller reacts at each turn, adjusting based on immediate obstacles. MPC, on the other hand, analyzes the entire maze, predicting potential paths and choosing the one that avoids dead ends and reaches the goal most efficiently.

Advantages of Model Predictive Control (MPC)

1. **Explicit Constraint Handling:** Unlike traditional control methods like PID, MPC can seamlessly incorporate operational constraints like maximum temperatures, pressure limits, or safety boundaries into its optimization problem. This ensures safe and feasible control actions at all times, reducing the risk of equipment damage or production disruptions.[1]
2. **Multi-objective Optimization:** MPC can optimize multiple objectives simultaneously, such as minimizing energy consumption, maximizing yield, and maintaining product quality. This allows for proactive control strategies that balance various priorities and improve overall process performance. [2]
3. **Improved Disturbance Rejection:** MPC's predictive nature allows it to anticipate and adjust to disturbances like changes in raw materials, feed rates, or environmental conditions. This proactive approach minimizes their impact on the process, leading to smoother operation and increased product consistency.
4. **Optimal Trajectory Planning:** MPC plans control trajectories over a defined prediction horizon, minimizing errors and optimizing performance throughout the entire sequence. This proactive approach leads to smoother transitions between operating points and minimizes overshoot or undershoot of control variables.[3]
5. **Potential for Integration with Advanced Technologies:** MPC readily integrates with machine learning, data analytics, and advanced optimization algorithms. This opens doors for further improvements in model accuracy, self-tuning capabilities, and optimal decision-making in complex operational environments.

Disadvantages of Model Predictive Control (MPC)

1. **Increased Computational Complexity:** Compared to simpler control methods like PID, MPC's optimization calculations can be more demanding and require sufficient computational resources. This can limit its application in real-time control of fast-paced or resource-constrained systems.[4]
2. **Dependence on Accurate Process Models:** The effectiveness of MPC relies heavily on the accuracy of the underlying mathematical model representing the

process dynamics. Inaccurate models can lead to suboptimal control performance or even instability. Developing and maintaining accurate models can be challenging and resource intensive.[5]

3. **Tuning Challenges**: Implementing and fine-tuning an MPC controller can be complex and require specific expertise in control theory and optimization algorithms. Choosing appropriate weights for different objectives and handling potential constraints can be a challenging optimization task.
4. **Potential for Over-optimization:** In some cases, aggressive optimization goals within MPC can lead to unnecessarily tight control actions and increased wear and tear on equipment. Balancing process performance with equipment longevity requires careful consideration of control objectives and constraints.

How Model Predictive Control Works

1. **Process Model and Prediction:**
   - MPC relies on a mathematical model that captures the dynamics of the controlled process. This model could be based on first-principles, data-driven techniques, or a combination of both.[6]
   - At each control step, the model is used to predict the future behavior of the process for a defined prediction horizon.
2. **Optimization with Constraints:**
   - Within the prediction horizon, MPC formulates an optimization problem that considers multiple objectives, such as minimizing energy consumption, maximizing yield, and maintaining product quality.
   - Operational constraints, like temperature limits, pressure boundaries, and safety requirements, are explicitly incorporated into the optimization problem to ensure safe and feasible operation.
3. **Optimal Control Trajectory:**
   - The optimization problem is solved to determine the optimal sequence of control actions over the prediction horizon that best satisfies the defined objectives and adheres to all constraints.
4. **Receding Horizon Implementation:**
   - Only the first control action in the calculated sequence is actually implemented on the process.[6]
   - As time progresses and new information becomes available, the prediction horizon is shifted forward (receding horizon), and the optimization problem is re-solved with updated data, leading to a refined control action for the next step.
5. **Continuous Feedback and Adjustment:**
   - MPC operates in a closed-loop fashion, continuously receiving feedback from sensors monitoring the actual process output.[7]

Applications of MPC:

Recent developments have seen applications expand across various engineering domains. The following highlights main movements:

1. **Process industry**
   Driven by the intricate nature of its processes, characterized by multiple variables and inherent time delays, MPC swiftly gained traction. The rapid adoption can be attributed to the economic advantages brought about by optimal control, particularly in optimizing large throughput. Here are key aspects of MPC applications in industrial processes:
   - **Complex Process Dynamics:**
     Model Predictive Control (MPC) demonstrably excels in applications involving complex multi-variable processes with significant time delays. Industries rife with intricate and interconnected systems, such as petrochemicals, chemical processing, and manufacturing, find substantial value in the predictive capabilities and optimized control trajectories offered by MPC.
   - **Energy Efficiency and Sustainability**:
     contributes to energy efficiency by optimizing the use of resources in industrial processes. This not only leads to cost savings but also aligns with sustainability goals.
2. **Robotics:**
   Mobile robots, industrial manipulators, and autonomous vehicles leverage MPC for precise trajectory tracking, obstacle avoidance, and optimal energy consumption.
3. **Power Electronics**
   Inverters and converters in renewable energy systems and electric vehicles employ MPC for efficient power generation and grid integration.

Introduction to SUMO robot

In the thrilling world of robotics, miniature gladiators clash in the arena of sumo robotics. These compact champions, known as sumo robots, are engineered marvels packing incredible force and agility into bite-sized packages. Their mission is to dominate the ring, pushing their opponents out of bounds in epic battles of strategy and power. [8]

At the heart of every sumo robot lies a carefully chosen arsenal of hardware components, each playing a crucial role in its quest for victory:

1. **Chassis:** The robot's foundation, built for strength and stability as shown Figure 1.

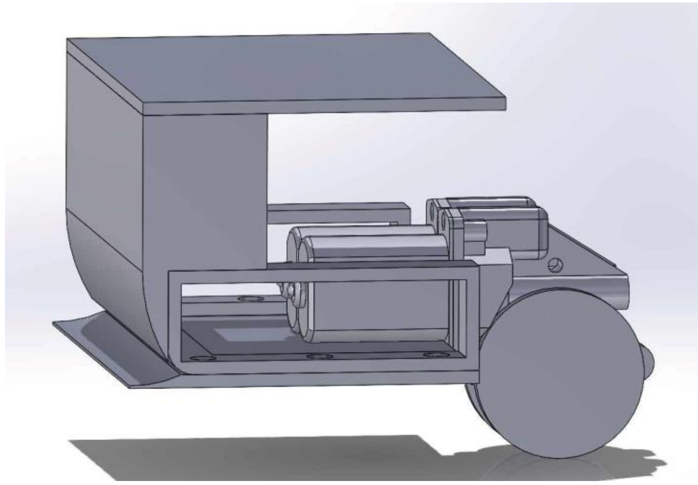

Figure 1: SUMO robot chassis

2. **Motors:** The driving force behind every maneuver.

   We use in our project DC motor.

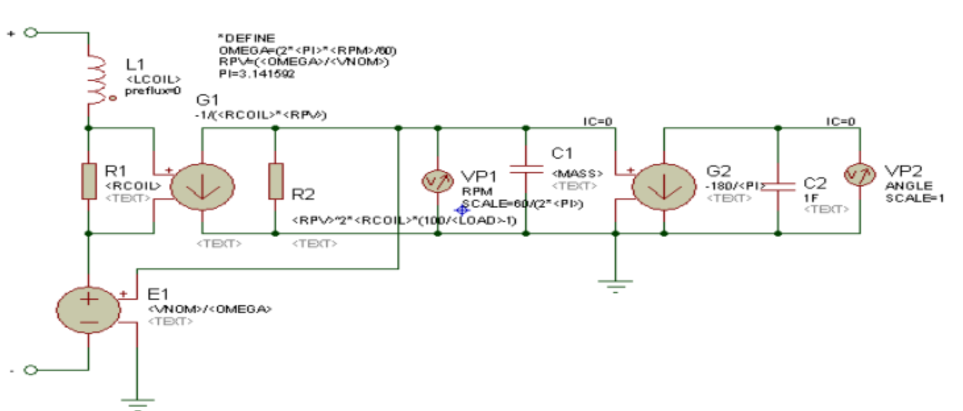

Figure 2 DC motor schematic model in Proteus

3. **Sensors**: The robot's eyes, we select IR sensor
4. **Microcontroller:** The robot's brain. This compact powerhouse processes sensor data, executes control algorithms, and sends commands to the motors, orchestrating the robot's actions and reactions in real-time.
5. **Battery:** The robot's lifeblood. A high-capacity battery fuels the fight.

Process Selection

We choose two 5V DC motors with encoder to control their speed separately with different required speeds for each and that will lead to change the direction of robot. The required speed depends on 3 infrared sensors for the robot as shown in Table 1

Table 1: The sensors and the motors relationship

Cases	Left sensor (L)	Center sensor (C)	Right sensor (R)	Output of left motor	Speed of left (RPM)	Output of Right motor	Speed of Right (RPM)
1	0	0	0	0	0	0	0
2	0	0	1	1	50	0	0
3	0	1	0	1	100	1	100
4	0	1	1	1	75	0	0
5	1	0	0	0	0	1	50
6	1	0	1	1	50	1	50
7	1	1	0	0	0	1	75
8	1	1	1	1	75	1	75

The cases are implanting using logical expression as shown in Table 2 and represent in MATLAB as shown in Figure 3:

Table 2: Logical expressions for required speeds

Speed (RPM)	**Logical expression**	**Logical expression**
50	C'.R	L.C'
75	C.R	L.C
100	L'.C.R'	L'.C.R'

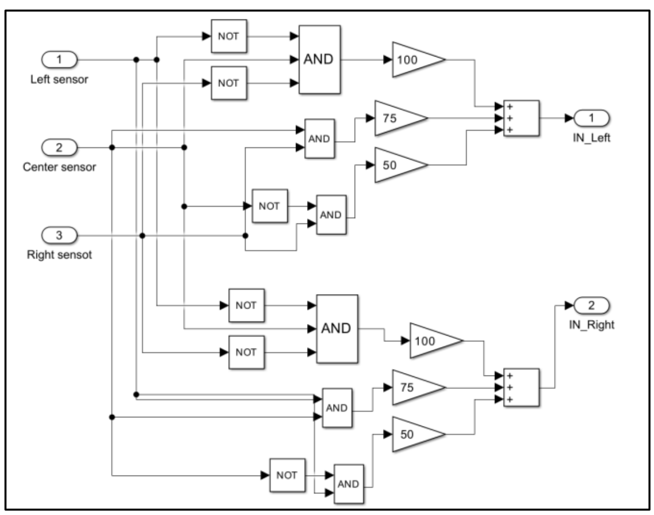

Figure 3:Logical expressions using MATLAB

DC motor modeling

For a dc motor open-loop dynamic equation is:

$$L\frac{di}{dt} + iR + k_b\overline{\omega} = V$$

$$J\frac{d\omega}{dt} = k_t i + T_d$$

where L is armature inductance (h), R is armature resistance (ohm), V is input voltage (volt), I for armature current (ampere), kb for back emf constants, and Kt reveals as torque constants.

In order to get step response data sample, DC motor model is feed with 15 V step input and then data is sampled in 0.25 second. Table 3 shown this sampling result.

Table 3: Speed of motor at 15 v step signal

Time, s	N, rpm	No.	N, rpm	No.	N, rpm	No.	N, rpm	No.	N, rpm
0	0	1	6.4900	2	8.8200	3	9.7100	4	9.99
0.25	0.6900	1.25	6.8200	2.25	8.9600	3.25	9.7600	4.25	9.99
0.5	1.5900	1.5	7.1100	2.5	9.0800	3.5	9.7900	4.5	9.99
0.75	2.3400	1.75	7.3600	2.75	9.2000	3.75	9.8100	4.75	10

Data in Table 3 is used for transfer function estimation. Transfer function estimation obtained using MATLAB, through the identification tool application. The model used is a process model for a second order system, without delay and overdamped. The transfer function estimation results are:

$$Estimated\ transfer\ function = \frac{0.676}{0.583s^2 + 2.629s + 1}$$

Suitability of process system for MPC

In the electrifying world of sumo robotics, victory hinges on a delicate balance of brute force and calculated movement. **While robust hardware lays the foundation, it's the strategic choreography of control that separates champions from contenders.** This is where Model Predictive Control (MPC) emerges as a game-changer for our sumo robot, granting it a decisive edge by anticipating opponent moves, optimizing trajectories, and adapting to the ever-shifting battlefield.

A Linear System Empowered: our sumo robot, though modeled as a linear system, can be transformed into a tactical mastermind with MPC. **Here's how:**

- **Precision Strikes and Agile Dodges:** MPC predicts your opponent's movements and your robot's dynamics, crafting optimal control sequences for powerful pushes, strategic evasions, and lightning-fast maneuvers. This proactive approach maximizes force delivery and minimizes wasted motion, leaving your opponent reeling.
- **Real-time Adaptability:** Sensors feed real-time information about the opponent's position and force into the MPC loop. This allows your robot to instantly adjust its strategies, countering unexpected moves and capitalizing on fleeting opportunities for domination.
- **Constraint Control:** Weight limits and joint range limitations are seamlessly integrated into MPC's optimization problem. This ensures safe and feasible control actions, protecting your robot from damage and optimizing energy expenditure for sustained combat power.
- **Disturbance Rejection:** Unpredictable bumps or uneven terrain are no match for MPC. The model predicts and compensates for external disturbances, maintaining your robot's balance and stability even in chaotic situations.

Beyond the Linear Domain: While your robot may be linear, MPC's versatility knows no bounds. Here's how it can **tackle non-linear challenges:**

- **Multi-model MPC**: This approach employs multiple models, each capturing different operating regimes of the non-linear system. MPC selects the most appropriate model based on real-time data, ensuring effective control across a broader range of conditions.
- **Advanced Optimization Techniques**: Nonlinear programming algorithms like quadratic programming can handle non-linear constraints and objective functions within the MPC framework, further expanding its control capabilities.

Design and Simulation

The parameters of MPC controller are:

1- Sample time T_s: using the open loop response in Figure 4 the rise time T_r almost equal to 5.287 sec, the sample time can be estimated to be $\frac{T_r}{20} \leq T_s \leq \frac{T_r}{10}$,so it is estimated to be $T_s = 0.5\ sec$.
2- Prediction horizon: using the open loop response in Figure 4 the settling time $T_{settling}$ almost equal to 10.402 sec, the prediction horizon can be estimated to be $\text{Prediction horizon} \geq T_{settling}$,so it is estimated to be $\text{Prediction horizon} = 11$.
3- Control horizon: the value of 3 is the most suitable value by tuning.

4- Constrains: the motor is working between 0RPM to 100RPM, so there is an output constrain of the MPC.
5- Weights: the is no constrain about the weights in our model.

6- Cost function:

$$J = \Sigma\,[r(k+i) - y(k+i)]^2 * Q + \Sigma\,\Delta u(k+i)^2 * R, \qquad i = 1 \text{ to } N$$

- Error Term: Penalizes the difference between desired output (r) and predicted output (y) over the prediction horizon (N).
- Control Effort Term: Penalizes excessive changes in control input (Δu) to promote smoother control actions.
- Weights (Q and R): Balance the relative importance of tracking error and control effort.

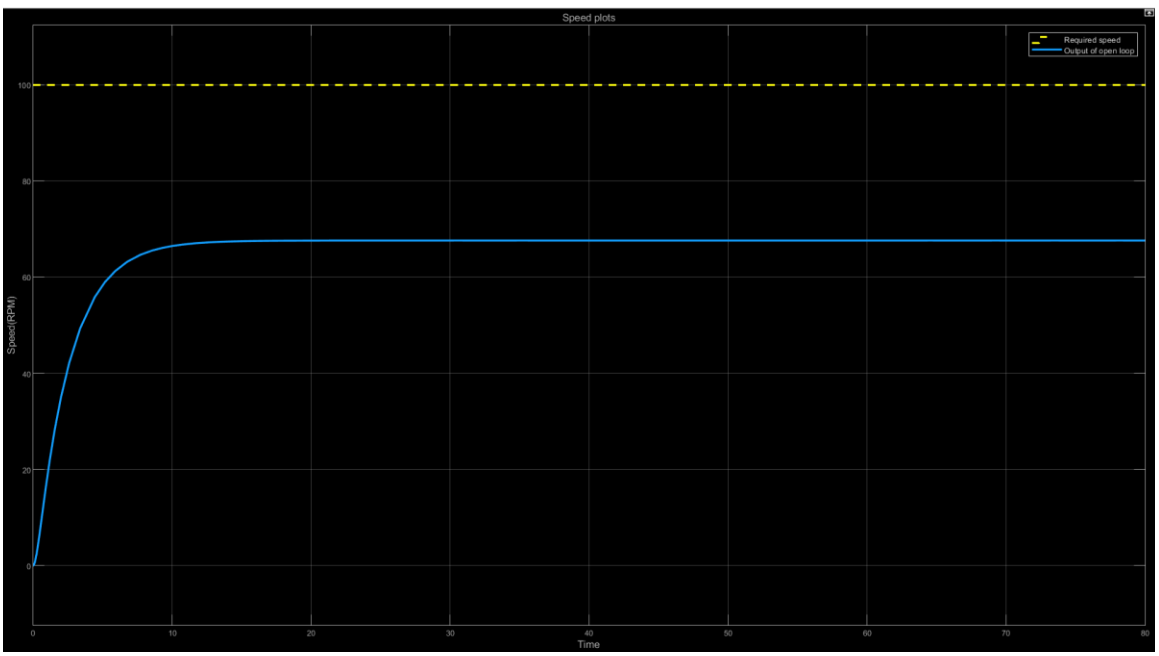

Figure 4: Response of open loop transfer function

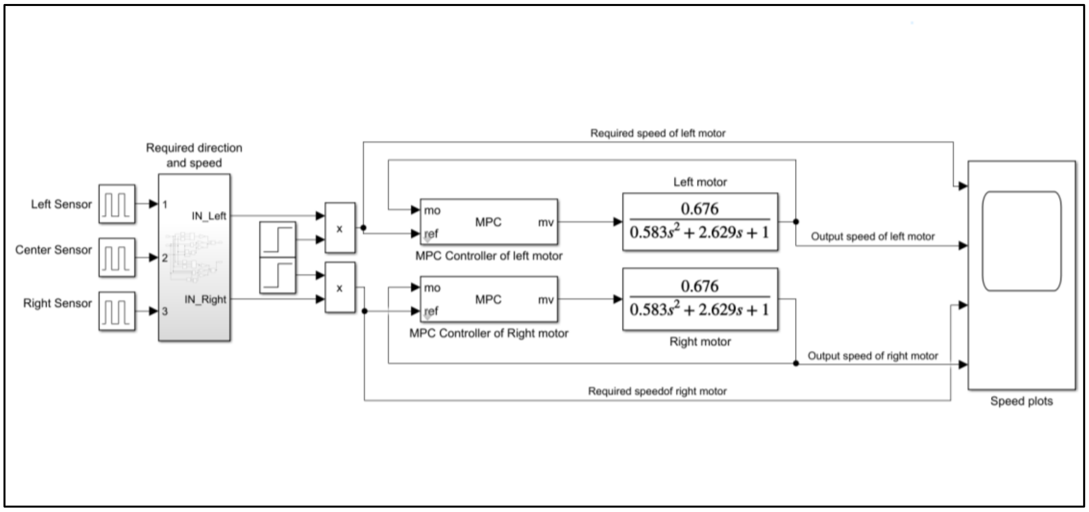

Figure 5: Diagram of MPC without disturbance

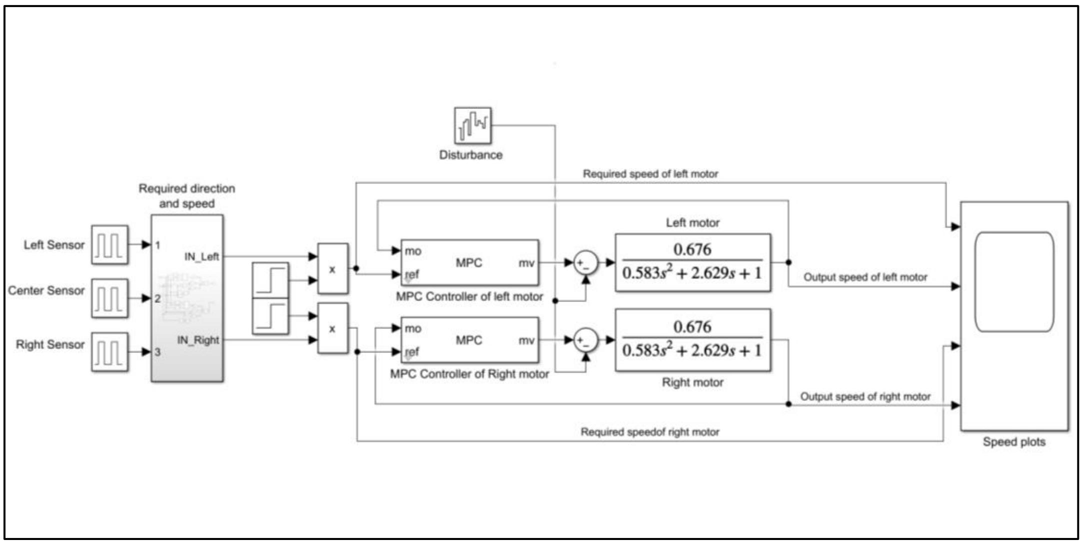

Figure 6: Diagram of MPC with disturbance

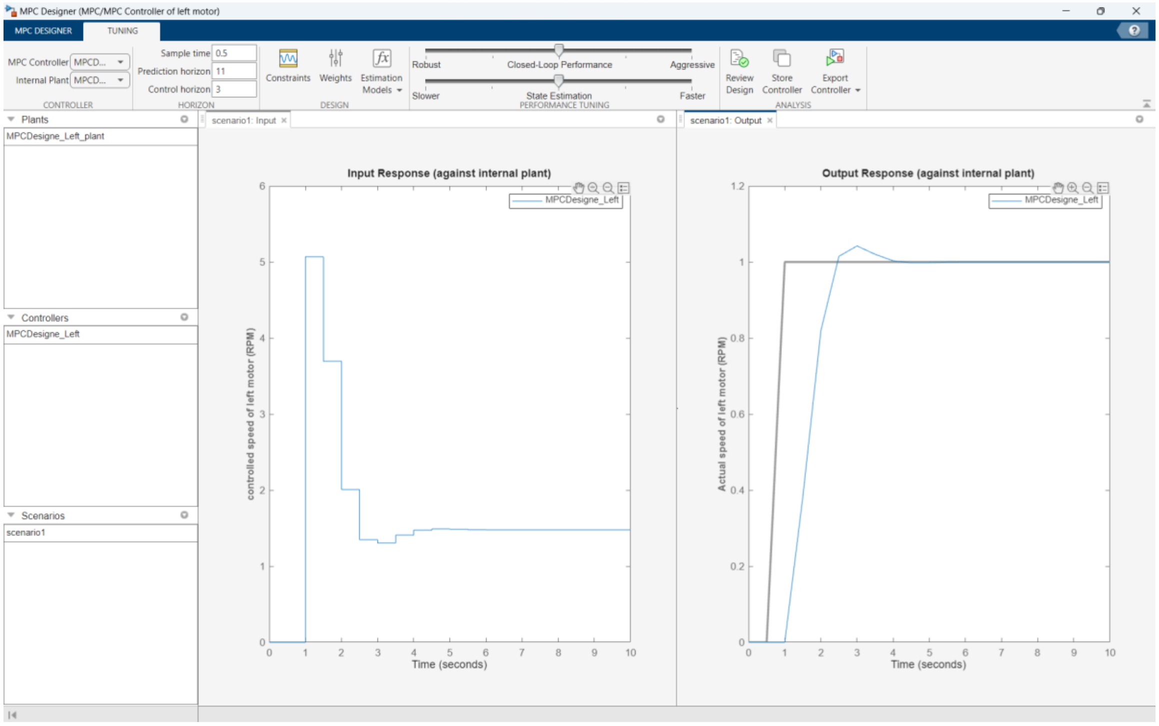

Figure 7: MPC designer for left motor

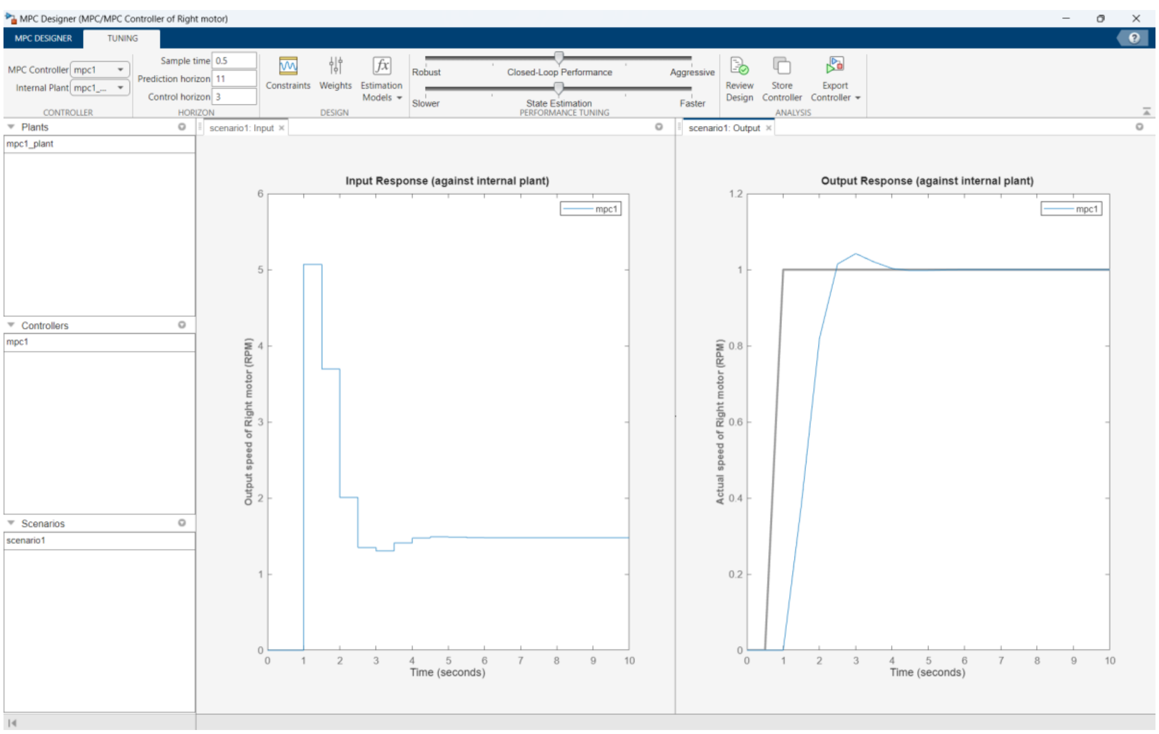

Figure 8: MPC designer for Right motor

Performance Analysis and Optimization

PID controller performance

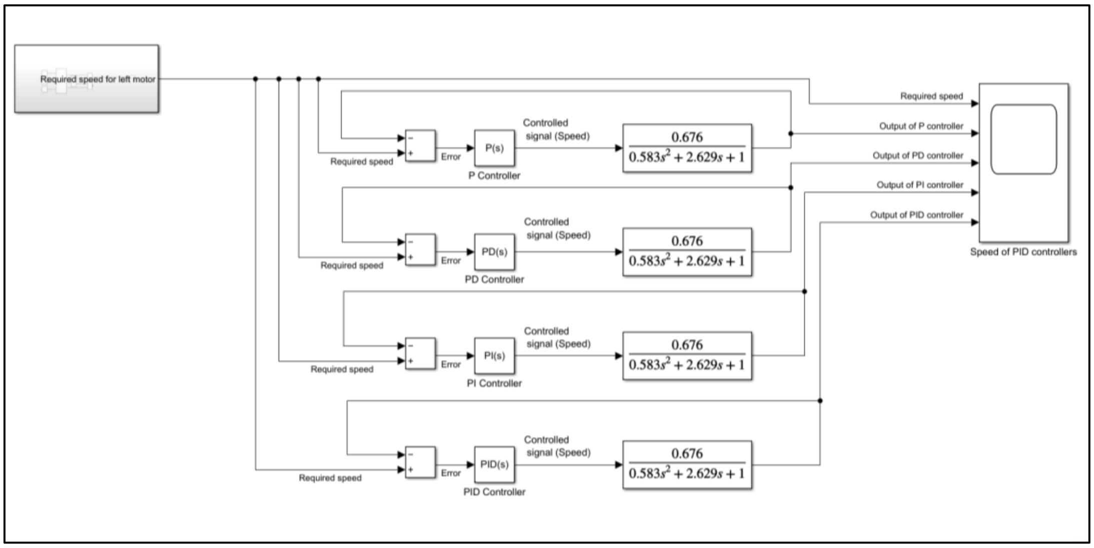

Figure 9: Diagram of PID without disturbance

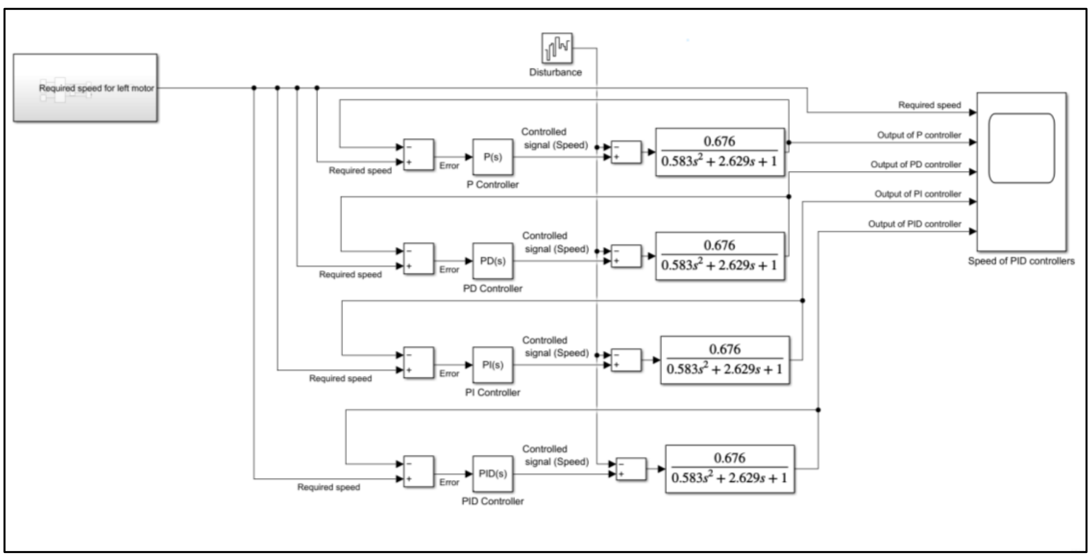

Figure 10: Diagram of PID with disturbance

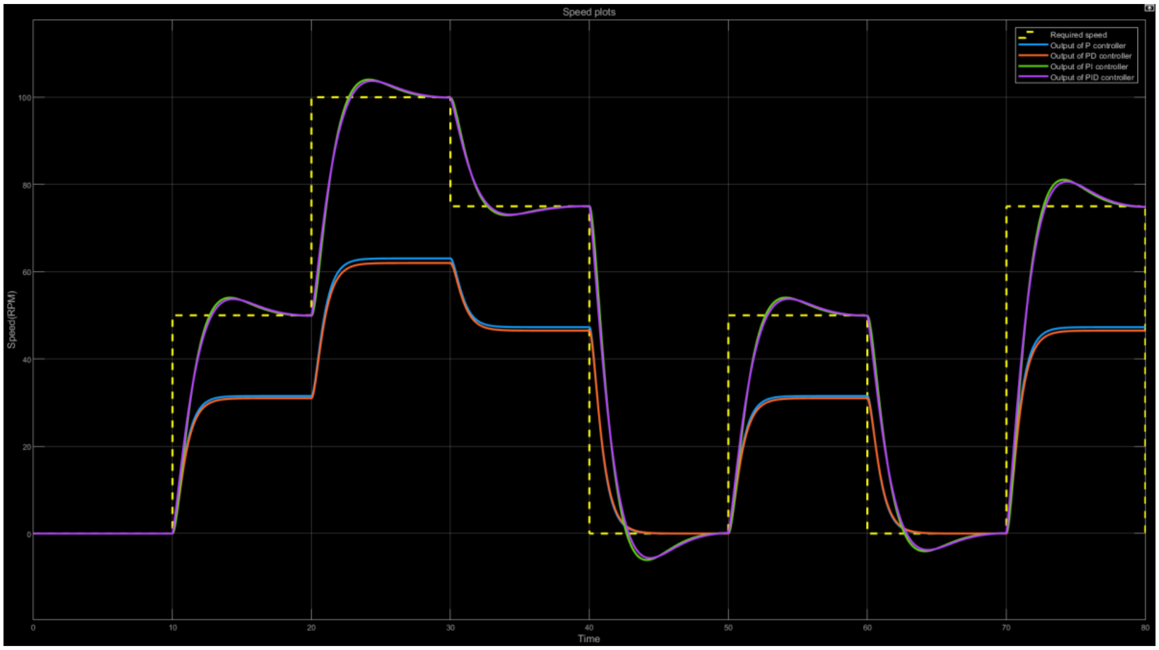

Figure 11: Responses of different PID controllers without disturbance

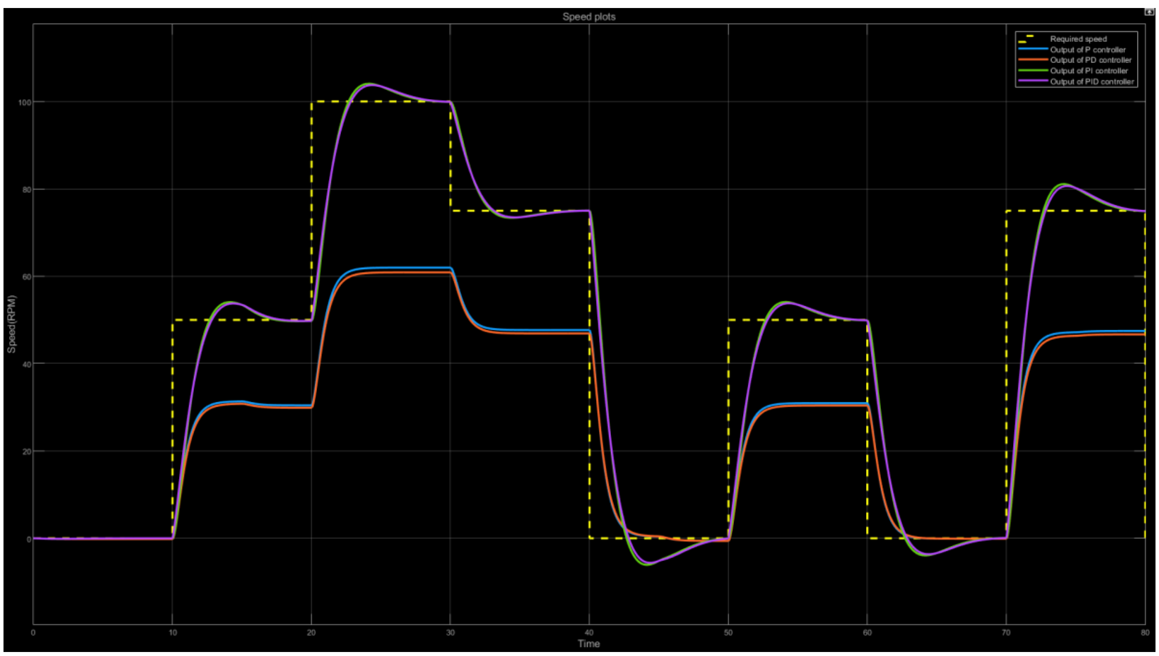

Figure 12: Responses of different PID controllers with disturbance

Proportional-Integral-Derivative (PID) controllers are widely used in control systems to regulate processes and maintain desired conditions. Each component in the PID controller (P, I, D) contributes to the overall control action, and their combination allows for versatile control in handling disturbances, following setpoints, and respecting constraints.

As shown in Figure 11 the responses of different types of PID controller without disturbance and it is clear that the PD and the PID controllers have almost zero offset and follows the setpoints. Also, there is no constrains that can apply to the responses of all types of PID controllers that means that if something goes wrong the PID controllers may lead to speed greater than the motor can handle.

There is a disturbance applies on the responses as shown in Figure 12 and it is applied every 15s and the physical meaning of this disturbance happen when the robot in attacking mode and there is a direct contact with the opponent the speed will decrease for small amount of time. The PID controllers here effected in different ways, but they are handling the disturbances with more time, so they have greater settling time to follow the setpoints.

MPC controller performance

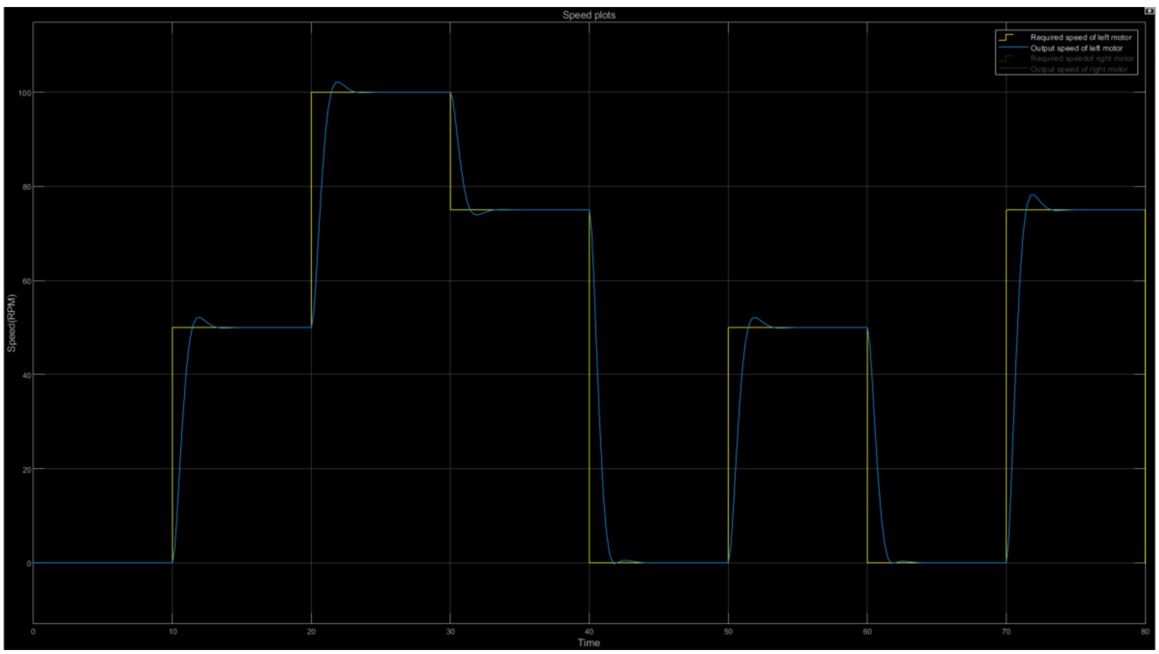

Figure 13: Left motor response of MPC without disturbance

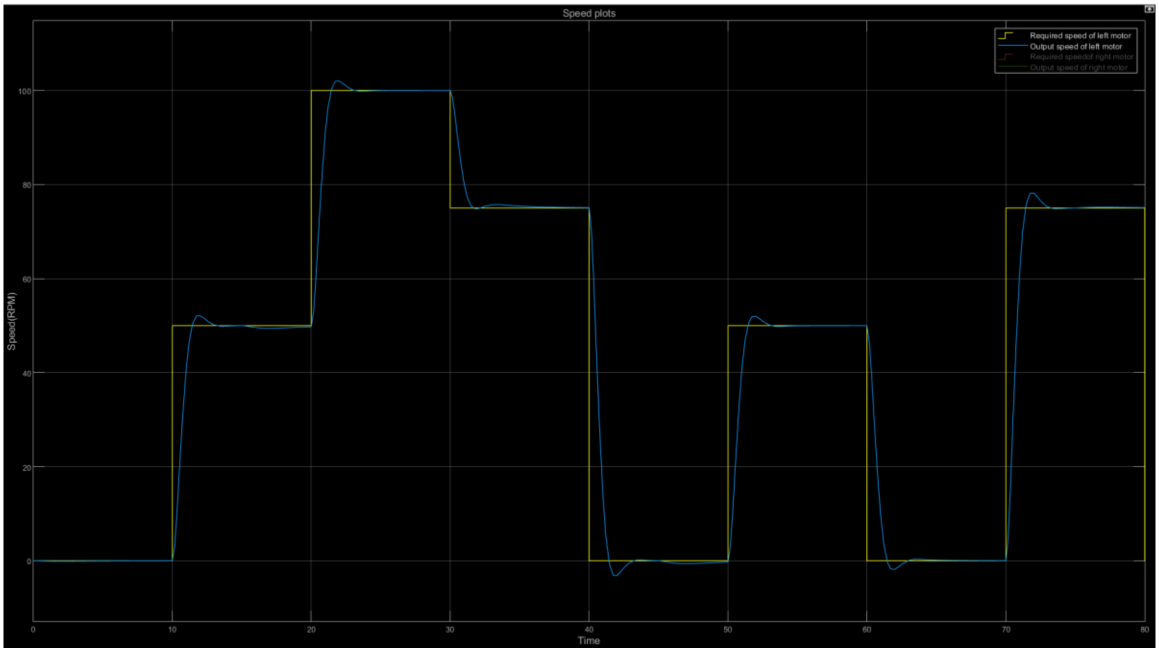

Figure 14 : Left motor response of MPC with disturbance

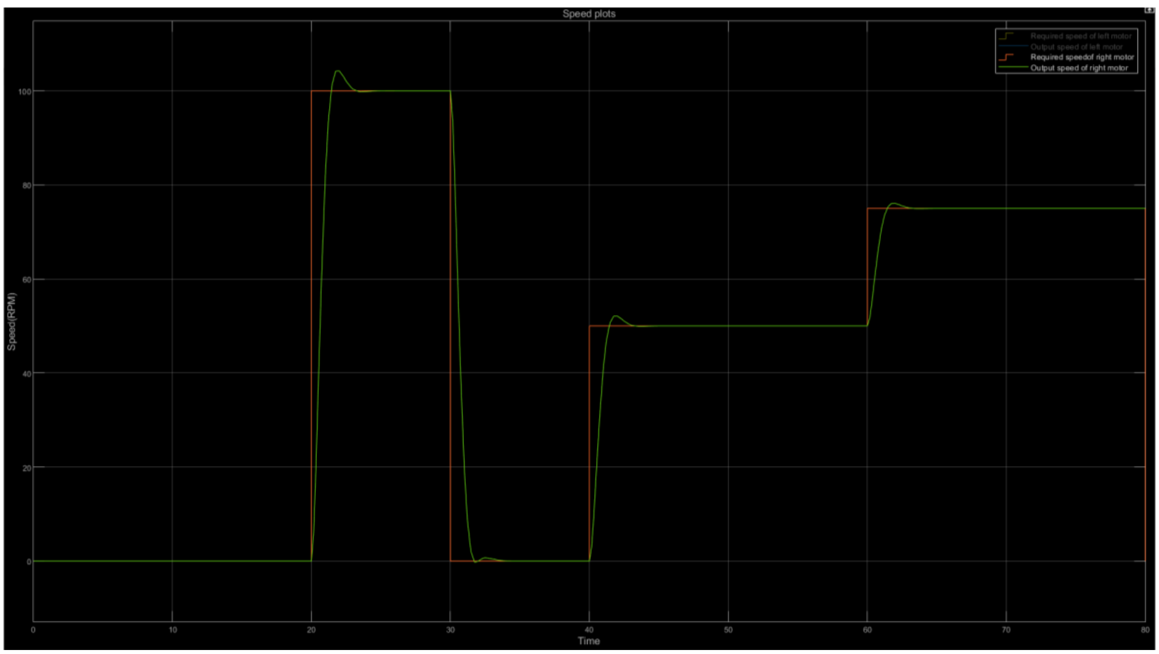

Figure 15: Right motor response of MPC without disturbance

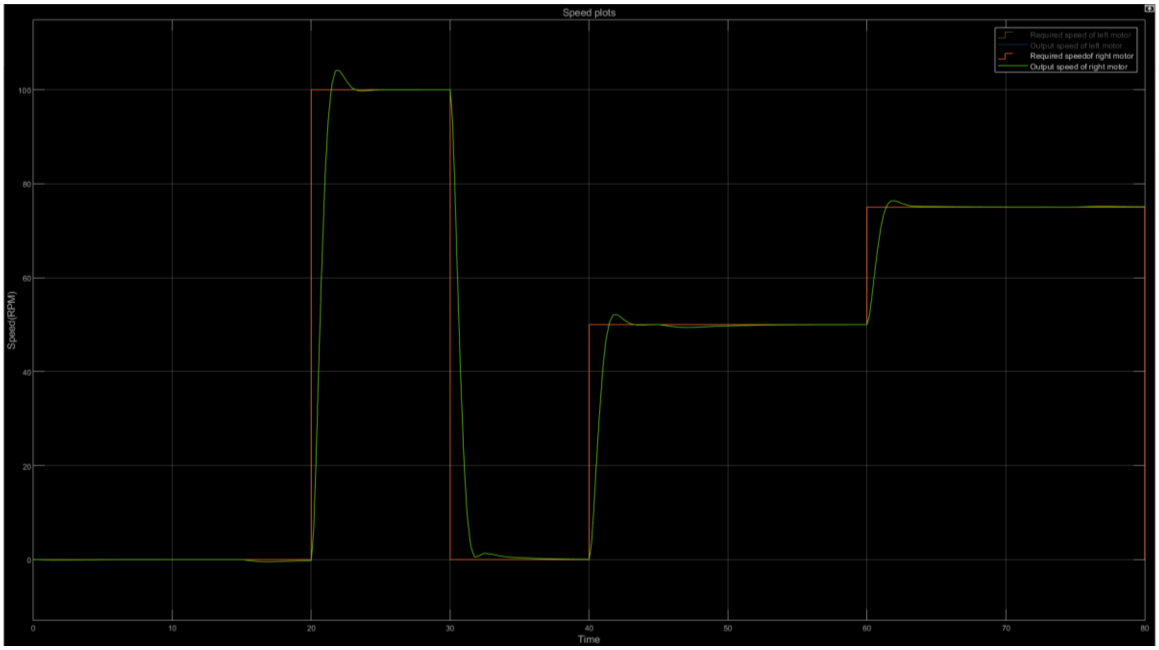

Figure 16: Right motor response of MPC with disturbance

As shown in Figure 13 and Figure 15 the responses of MPC controller without disturbance and it is almost zero offset and follows the setpoints. Also, there is constrains that can apply here like the maximum speed is 120 RPM for 100 RPM to the responses.

There is a disturbance applies on the responses as shown in Figure 14 and Figure 16 and it is applied every 15s and the physical meaning of this disturbance happen when the robot in attacking mode and there is a direct contact with the opponent the speed will decrease for small amount of time. The MPC controller here almost doesn't effected and handling the disturbances faster.

Comparisons between controllers

- <u>Handling Disturbances:</u>
    1. Proportional (P): It provides a control output that is proportional to the current error. It helps in reducing the steady-state error caused by disturbances but may not eliminate it entirely.
    2. Integral (I): Accumulates the past error over time and acts to eliminate any residual steady-state error. It is effective in handling disturbances that persist over time.
    3. Derivative (D): Responds to the rate of change of the error. It helps to anticipate future behavior based on the current rate of change and is effective in reducing the impact of sudden disturbances.
    4. MPC: It incorporates a predictive model of the system, which allows it to anticipate the effect of disturbances over a prediction horizon. The optimization algorithm considers future disturbances in determining the optimal control sequence, enabling the controller to proactively mitigate the impact of disturbances.

- <u>Following Setpoints:</u>
    1. Proportional (P): Provides an immediate response to changes in setpoint, but may result in steady-state error.
    2. Integral (I): Eliminates steady-state error over time and ensures the system settles at the desired setpoint.
    3. Derivative (D): Damps oscillations caused by rapid changes in setpoint, improving stability.
    4. MPC: It is designed to track reference trajectories or setpoints by optimizing the future control actions. The objective function includes terms related to minimizing deviations from setpoints, ensuring that the system follows the desired trajectory.
- <u>Respecting Constraints:</u>
    1. Proportional (P): Can be limited by the proportional gain to prevent excessive control effort.
    2. Integral (I): Can lead to windup issues if not properly handled, where the integral term accumulates excessively during saturation.
    3. Derivative (D): Can contribute to control effort limitations by damping rapid changes.
    4. MPC: It naturally handles constraints on both control inputs and system states. The optimization problem includes constraints to ensure that the predicted behavior of the system adheres to specified limits. This makes MPC well-suited for systems with physical or operational constraints.

As shown in Table 4 there is a compression between all types of controllers that used for variate setpoints assuming there are no disturbances.

One of the best criteria to deiced what is the better controller is the mean squad error (MSE). The MSE of PI is 263.0083 and PID is 241.7129 that’s mean that the PID is the better controller here, but comparing with MPC, it is the best for our model.

Table 4: comparing the controllers

Parameter	P controller	PI controller	PD controller	PID controller
Settling time	3.408	7.305	3.473	7.805
Rise time	1.659	1.959	1.719	1.959
Maximum overshoot	0.504%	8.152%	0.504%	6.989%
Steady state error	36.80	0.001	37.81	0.003
Mean squad error	--	263.0083	--	241.7129

Challenges and insights

We have a challenge to replace the PID controller to the MPC and to analysis our response in the sumo robot.

Imagine a sumo robot, not battling another in the ring, but wrestling with a giant calculator. That's the image of our cost challenge. Every fancy MPC calculation comes with a financial jab. We need to find ways to optimize our control strategies, squeeze the most performance out of every penny, and maybe even discover some budget-friendly hacks along the way.

The Learning Curve: From Challenges to Growth.

But here's the flip side: even with the cost crunch, MPC is an investment in learning and growth. It's like a sensei throwing us into the training ring, pushing us to innovate and find creative solutions. We'll gain valuable experience in:

- Optimizing control strategies: Figuring out how to get the most out of our robot's movements with limited resources.
- Managing cost considerations: Becoming budget ninjas, squeezing every drop of performance from every penny.
- Implementing cutting-edge algorithms: Getting our hands dirty with the nitty-gritty of real-world MPC application.

Conclusion

Table 5: Comparisons

Feature	MPC	PID
Approach	Proactive, optimizes future behavior	Reactive, responds to errors in real-time
Constraint Handling	Excellent, can factor in obstacles and environment	Limited, may struggle with complex constraints
Performance Optimization	Yes, can optimize multiple objectives	Limited, primarily focuses on error reduction
Multi-Objective Control	Well-suited for complex maneuvers	Can be challenging with multiple actuators and sensors
Disturbance Rejection	Strong, can adapt to unexpected changes	Weaker, may struggle with significant disturbances

While PID controllers remain a valuable tool, MPC's advanced capabilities make it a promising option for sumo robot control. Its ability to handle constraints, optimize performance, and adapt to disturbances offers significant advantages in the competitive world of sumo robotics. As research and development progress, we can expect to see MPC play an increasingly crucial role in shaping the future of these miniature gladiators.

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